CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**RAILWAYS, AIRPORTS, DOCKS AND HARBOR ENGINEERING** **Professional Elective-IV**	L	T	P	C
23A01702b		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the components and geometric design principles of railway tracks.
2. Analyze the principles of railway track design, signaling, and interlocking.
3. Evaluate airport site selection, runway orientation, and terminal area planning.
4. Design runways and taxiways based on aircraft characteristics and geometric elements.
5. Assess the requirements and classification of ports, harbors, docks, and navigation aids.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Explain railway track components, functions, and requirements.
2. Apply geometric design principles to railway track layout and interlocking systems.
3. Evaluate airport planning aspects, including site selection, runway design, and terminal planning.
4. Design runways and taxiways based on geometric standards and safety regulations.
5. Analyze ports and harbor structures, including docks, breakwaters, and navigation aids.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	1	1	-	-	1	-	-	-	1	-	1	1	1
CO -2	3	2	2	1	2	-	-	-	-	-	-	1	2	1
CO -3	3	2	3	2	2	-	1	-	-	-	-	1	2	2
CO -4	3	1	2	2	3	1	-	-	-	-	-	1	2	2
CO -5	3	2	2	1	2	1	-	-	-	-	-	1	2	1

UNIT – I

Railway Engineering

Introduction – Permanent Way Components – Cross Section of Permanent Way – Functions and Requirements of Rails, Sleepers and Ballast – Types of Gauges – Creep of Rails – Theories Related to Creep – Coning of Wheels – Adzing of Sleepers – Rail Fastenings

UNIT – II

Geometric Design of Railway Track

Gradients – Grade Compensation – Cant and Negative Super Elevation – Cant Deficiency – Degree of Curves – Safe Speed On Railway Track – Points and Crossings – Layout and Functioning of Left Hand Turn Out and Right Hand Turn Outs – Station Yards – Signaling and Interlocking

UNIT – III

Airport Engineering

Airport Site Selection – Factors Affecting Site Selection and Surveys- Runway Orientation – Wind Rose Diagram – Basic Runway Length – Correction for Runway Length – Terminal Area – Layout and Functions – Concepts of Terminal Building – Simple Building , Linear Concept, Pier Concept and Satellite Concept – Typical Layouts

UNIT – IV

Geometric design of runways and taxiways

Aircraft Characteristics – Influence of Characteristics On Airport Planning and Design – Geometric Design Elements of Runway – Standards and Specifications - Functions of Taxiways – Taxiway Geometric Design – Geometric Elements and Standard Specifications – Runway and Taxiway Lighting.

UNIT – V

Ports and Harbors

Harbours - Requirements of Ports and Harbors – Types of Ports – Classification of Harbors – Docks and Types of Docks – Dry Docks, Wharves and Jetties – Breakwaters: Layouts of Different Types of Harbors and Docks – Dredging Operations – Navigation Aids

TEXT BOOKS:

1. A Text Book of Railway Engineering-S.C.Saxena andS.Arora, Dhanpatrai and Sons, New Delhi 2010
2. Highway, railway, Airport andHarbour Engineering – K.P. Subramanian, Scitech Publishers

REFRENCE BOOKS:

1. Harbour, Dock and Tunnel Engineering – R. Srinivasan, Charotar Publishing House Pvt. Limited, 200
2. Railway Track Engineering by J.S.MundreyMcGraw Hill Education 5th edition 2017
3. A Text book of Transportation Engineering – S.P.Chandola – S.Chand& Co. Ltd. – (2001)

Online Learning Resources:

https://nptel.ac.in/courses/105107123
https://archive.nptel.ac.in/courses/114/106/114106025/
https://www.mkube.co.in/product-detail/railways-airports-docks-and-harbour-engineering